NeurIPS Paper Checklist — AutoResearch on Midwest IT MSPs

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Companion to `FinalReport.pdf`. Section / Table / Appendix references below point into the main report.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Section 1 (Introduction) of the main report state three contributions — a SHA-256-locked Worker/Judge architecture with Decoupled Isolation protocols, a 20-experiment trajectory identifying two structural signals (`tenure_sq`, `rev_per_emp`) and one audit-detected artifact (Exp_016), and a single-pass holdout RMSE of 1.5119 below the pre-declared < 2.0 ceiling — and all three are substantiated by Sections 2–4 of the main report.

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2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 5 (Discussion) of the main report discusses (a) the upstream-filter trap of restricting to Midwest IT MSPs; (b) the $\sim 16\%$ tag-rate sparsity floor for binary NLP features at N=62 with its mathematical derivation; (c) the absence of inter-rater reliability on the 62 manual labels; (d) the agents-vs.-humans audit lesson surfaced by the Exp_016 *compliance* artifact; and (e) computational/cost scaling.

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   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Section 2 (Method) of the main report fully specifies the Worker/Judge pipeline (Ridge $\alpha{=}1.0$, `StandardScaler`, 11 features, `KFold(5, shuffle=True, random_state=42)`, clip to $[1.0, 10.0]$, 0.5-grid rounding) and the SHA-256 hash of the locked Judge. Appendix A (Reproducibility) of the main report names the two reproduction commands (`python final_test_eval.py` for the holdout table, `python run_experiment.py "..." -keep` for the training OOF). The 20 per-experiment snapshots in `logs/Snapshot_model_Exp_*.py` and the committed `scrape_cache.json` make every intermediate state recoverable.

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5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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Justification: The full repository (`model.py`, `eval/prepare.py`, `run_experiment.py`, `verify_integrity.py`, `final_test_eval.py`, all 20 Worker snapshots, both labeled CSVs, and the scraper cache) is available at GitHub. The README documents the reproduction commands and Appendix A of the main report names the exact entry points and the SHA-256 baselines for both the Worker champion and the Judge.

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   Answer: [Yes]

   Justification: Section 2 (Method) of the main report reports the train/test design (62-firm 5-fold shuffled KFold, seed 42, for development; 28-firm locked single-pass holdout for the final number), the Ridge α=1.0 baseline and the 5$\times$ Alpha Guardrail on hyperparameter loops, the `StandardScaler` preprocessor, and the 11-feature set; Table 2 of the main report reports both partition metrics.

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7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

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   Justification: We use an empirical cross-experiment noise band (~ 0.003 RMSE, measured on the Exp_006 rounding-only Control) as our effect-size yardstick; the train$\rightarrow$test gap of $+0.1164$ RMSE (Table 2 of the main report) exceeds this band by $\sim$ 40$\times$, and the Exp_018 ablation's $+0.097$ RMSE swing exceeds it by $\sim$ 32$\times$. Per-firm error distribution at the holdout stage is reported in Section 4 (Results) of the main report (89% within ± 2.0 points).

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Justification: Section 5 (Discussion) of the main report reports compute (single MacBook CPU, no GPU), aggregate wall-clock ($\sim$ 25 s for all 20 controlled experiments at warm cache), and the cumulative dollar cost (\$1.24, one-time Apollo firmographics). All 20 discards as well as the keeps are present in `logs/results.tsv`; no failed experiments were hidden from the trajectory.

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Justification: The released artifact is a 10-feature Ridge regression and a 62-row tabular label set; it is neither a generative model nor a high-risk scraped corpus, and poses no plausible dual-use risk requiring access controls.

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    Justification: The 62 Manual Score labels were produced by the sole author drawing on the Stanford/Yale search-fund primers; no crowdsourcing platform or external human subjects were involved.

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    Answer: [Yes]

    Justification: The AutoResearch loop — hypothesis generation, Worker code edits across all 20 controlled experiments, per-experiment log writing, and diagnostic interpretation — was driven by Claude Code (Anthropic) under the contract specified in `program.md`. The SHA-256-locked Judge enforced an inviolable evaluation boundary throughout, and the author retained final *keep*/*discard* approval. Declared in Section 2 (Method) of the main report and the Acknowledgments.

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